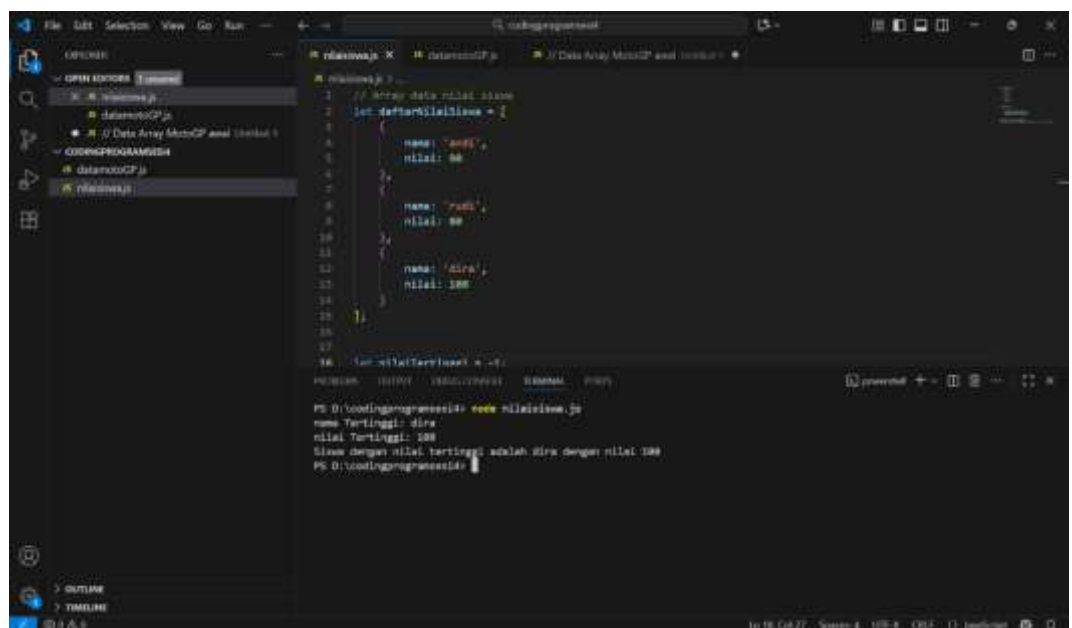


TUGAS 4 OBJECT LITERAL

1. Terdapat objek yang menyimpan data nilai siswa seperti berikut ini:

```
let studentsScore = [  
  {  
    name : 'Andi',  
    score: 90  
  },  
  {  
    name : 'Rudi',  
    score: 80  
  },  
  {  
    name : 'Dira',  
    score: 100  
  },  
]
```

Tampilkan hasil output berupa nama dan nilai siswa yang paling besar !



The screenshot shows a code editor with a file explorer on the left and a terminal at the bottom. The code in the editor defines an array of student scores and uses a loop to find the student with the highest score.

```
1 // Array data nilai siswa  
2 let daftarNilaiSiswa = [  
3   {  
4     name: 'Andi',  
5     nilai: 90  
6   },  
7   {  
8     name: 'Rudi',  
9     nilai: 80  
10  },  
11  {  
12    name: 'Dira',  
13    nilai: 100  
14  },  
15 ]  
16  
17 let nilaiTertinggi = -1;  
18  
19 for (let i = 0; i < daftarNilaiSiswa.length; i++) {  
20   if (daftarNilaiSiswa[i].nilai > nilaiTertinggi) {  
21     nilaiTertinggi = daftarNilaiSiswa[i].nilai;  
22     namaTertinggi = daftarNilaiSiswa[i].name;  
23   }  
24 }  
25  
26 console.log(`Nama dengan nilai tertinggi adalah ${namaTertinggi} dengan nilai ${nilaiTertinggi}`);
```

The terminal output shows the result of the code execution:

```
PS D:\coding\programasid> node nilaiSiswa.js  
Nama dengan nilai tertinggi adalah Dira dengan nilai 100  
PS D:\coding\programasid>
```

TUGAS

2. Terdapat Array of objek yang menyimpan data-data event MotoGP seperti berikut ini :

```
var motoGP = [  
  {  
    circuit: 'Losail',  
    location: 'Qatar',  
    winner: {  
      firstName: 'Andrea',  
      lastName: 'Dovizioso',  
      country: 'Italy'  
    }  
  },  
  {  
    circuit: 'Autodromo',  
    location: 'Argentina',  
    winner: {  
      firstName: 'Cal',  
      lastName: 'Crutchlow',  
      country: 'UK'  
    }  
  },  
  {  
    circuit: 'De Jerez',  
    location: 'Spain',  
    winner: {  
      firstName: 'Valentino',  
      lastName: 'Rossi',  
      country: 'Italy'  
    }  
  },  
  {  
    circuit: 'Mugello',  
    location: 'Italy',  
    winner: {  
      firstName: 'Andrea',  
      lastName: 'Dovizioso',  
      country: 'Italy'  
    }  
  }  
]
```

Kelompokkan data tersebut berdasarkan Negara Pemenang, sehingga hasil outputnya seperti berikut ini:

```

{
  Italy: {
    winningCircuits: [
      { name: 'Andrea Dovizioso', winLocation: 'Isle of Man, Qatar' },
      { name: 'Valentino Rossi', winLocation: 'De Jerez, Spain' },
      { name: 'Andrea Dovizioso', winLocation: 'Mugello, Italy' }
    ],
    totalWins: 3
  },
  UK: {
    winningCircuits: [
      { name: 'Cal Crutchlow', winLocation: 'Autodromo, Argentina' }
    ],
    totalWins: 1
  }
}

```

```

// data.js
const races = [
  { raceId: 1, winner: 'Andrea Dovizioso', location: 'Isle of Man', circuit: 'Qatar' },
  { raceId: 2, winner: 'Valentino Rossi', location: 'De Jerez', circuit: 'Spain' },
  { raceId: 3, winner: 'Andrea Dovizioso', location: 'Mugello', circuit: 'Italy' },
  { raceId: 4, winner: 'Cal Crutchlow', location: 'Autodromo', circuit: 'Argentina' }
];

function groupRecordsByWinnerCountry(data) {
  return data.reduce((acc, currentRace) => {
    const country = currentRace.winner.country;

    const raceInfo = {
      name: `${currentRace.winner.firstName} ${currentRace.winner.lastName}`,
      winLocation: `${currentRace.location}, ${currentRace.circuit}`
    };

    if (!acc[country]) {
      acc[country] = {
        winningCircuits: [raceInfo],
        totalWins: 1
      };
    }
  }, {});
}

// console.log(groupRecordsByWinnerCountry(races));

```

Problem: OUTPUT: Results: console.log(groupRecordsByWinnerCountry(races));
 {
 Italy: { winningCircuits: [{name: 'Andrea Dovizioso', winLocation: 'Mugello, Italy'}, {name: 'Andrea Dovizioso', winLocation: 'Isle of Man, Qatar'}, {name: 'Valentino Rossi', winLocation: 'De Jerez, Spain'}], totalWins: 3 },
 UK: { winningCircuits: [{name: 'Cal Crutchlow', winLocation: 'Autodromo, Argentina'}], totalWins: 1 }
 }
 console.log(groupRecordsByWinnerCountry(races));